REFLECTION DRAFT - REWRITE THIS BY HAND BEFORE SUBMISSION

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Cloud Computing Mini Project: Command Parser

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Repository: github.com/shobhit2294/230101046_assignment_cloud_computing

What I learned

This project helped me understand sequence-level knowledge distillation. The smaller Pythia model learns from text generated by the Phi-3.5 teacher; it does not receive the teacher weights or logits. Since the teacher runs through Ollama and returns text, the training setup uses those generated examples as its targets.

I also learned how the pieces of a small cloud deployment fit together. SeaweedFS provides S3-compatible object storage, while k3d runs the Kubernetes deployment locally. Readiness checks, resource limits, rolling updates, and an HPA help make the service easier to operate, although a single-node cluster is not highly available.

The networking exercise showed why a network ACL needs explicit return-traffic rules: it is stateless, and the first matching rule determines the result. The sample NACL passed all six supplied simulator checks.

Challenges and results

Cleaning the teacher output required explicit validation and duplicate removal. Of 3,000 input commands, 2,861 were kept. The trained student produced valid JSON for 99% of the validation examples and achieved 43% full-frame accuracy, compared with 48% for the teacher. This shows that the smaller model learned much of the task but still made mistakes.

The cluster load test measured a 3,300 ms p95 latency and 2.41 requests per second. Three cold-start runs each took 8 seconds. The estimated cost was 0.91 INR per 1,000 requests, based on the stated two-replica resource assumptions.

Reliability result to discuss honestly

The recorded rolling-update test had 2 non-200 responses out of 39 requests (5.13%). That is above the runbook limit of 0.5%, so this check did not pass and needs a successful rerun. In the separate steady-state load test, 211 of 213 requests succeeded; the two errors were HTTP 500 responses for invalid model output.

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What I would improve

I would add request batching to improve CPU inference throughput and use a shared work queue so that one slow labeling shard does not hold up the whole job. I would also test the deployment on multiple nodes and repeat the rolling-update test until it meets the stated failure limit. A larger student model or cleaner training labels could improve full-frame accuracy.

How AI assistance was used

An AI coding assistant helped draft parts of the code and project documentation. The repository records this use in AI_USE.md. The project includes checker and test scripts, and the results quoted here are taken from the saved project evidence. Before submitting, I will make sure this description matches what I personally ran, checked, and learned.

Before submitting

Rewrite this reflection in your own words and handwriting. Remove or change anything that does not describe your experience. Use only results you can explain. Scan the handwritten pages and save the scan as report/230101046.pdf. Do not submit this typed draft as the handwritten reflection.